METRON: Metabolic Dynamic Perception Kolmogorov-Arnold Network for Biological Age Estimation

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***Abstract*—Biological age is a more direct reflection of physiological status than chronological age, serving as a vital measure to evaluate health risks and aging interventions. While steroid metabolomics offers rich information for exploring aging mechanisms, the complex and nonlinear interactions within metabolic networks remain challenging in modeling. Here, we propose and describe METRON as a deep learning framework to predict biological ages from steroid metabolomics. Specifically, a Metabolite Interaction Perception Module (MIPM) is proposed to capture the interactions. Subsequently, a Group-Rational Kolmogorov-Arnold Network is also integrated to capture intricate dependencies and enhance the representation capability. We demonstrate that METRON achieves promising performance as compared to other machine learning and deep learning methods. Beyond performance, METRON offers interpretability by recovering the established markers such as Dehydroepiandrosterone (DHEA) and identifying 17-hydroxyprogesterone (17-OH-P4) as the key signature linked to hypothalamic-pituitary-adrenal axis dynamics. These results support the capacity of METRON not only to estimate biological age but also to uncover underappreciated metabolic drivers behind aging.**

***Index Terms*—Deep learning, biological age prediction, metabolomics, Kolmogorov-Arnold Networks, deformable attention, interpretability.**

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I. Introduction

Aging is a complex and multifactorial process [1] characterized by the progressive accumulation of cellular and molecular damage, which leads to a decline in physiological function and an increased risk of age-related diseases [2]. Conditions such as Alzheimer's disease, Parkinson's disease, and osteoporosis are intrinsically linked to aging, contributing substantially to the health challenges faced by the global population [3]. Therefore, the development of biological age (BA) estimation, as a reflection of an individual's physiological state distinct from chronological age (CA), is a critical measurement for understanding the mechanisms of aging and developing effective interventions [4].

To achieve accurate BA estimation, research has shifted towards intrinsic molecular measures with omics technologies [5]. Among these, metabolomics has emerged as a powerful tool due to its dynamic sensitivity to an individual's biological state [6]. Specifically, steroid hormones, which regulate key metabolic processes, have been identified as promising indicators for refining BA prediction [7].

However, modeling the relationship between steroid metabolomics and biological age presents considerable computational challenges due to the complexity and nonlinear nature of metabolic networks. While recent advances have employed various machine learning approaches ranging from traditional regression to deep neural networks (DNNs) [8], [9], existing methods often suffer from limitations such as reliance on extensive multi-stage preprocessing, lack of interpretability, or poor generalization due to overfitting [10]. Furthermore, current state-of-the-art frameworks often require domain-specific weight initialization and separate models for different demographics, limiting their applicability [9].

To address these challenges, we introduce the Metabolic Dynamic Perception Kolmogorov-Arnold Network (METRON), an end-to-end framework for biological age prediction. METRON is designed to learn directly from raw metabolomics data, eliminating the need for complex and multi-stage preprocessing and domain-specific weight initialization. Furthermore, it operates as a single and unified model for both sexes while integrating demographic information to enhance predictive accuracy. The main contributions of this work are summarized as follows:

1) We propose METRON, an end-to-end deep learning framework for biological age prediction from metabolomics data. The architecture uniquely combines a Metabolite Interaction Perception Module (MIPM) to adaptively capture inter-metabolite relationships and a Group-Rational Kolmogorov-Arnold Network (GR-KAN) to model complex nonlinear patterns with high expressive ability.
2) We demonstrate through comparative experiments that METRON outperforms a wide range of benchmark methods. This includes classical machine learning models, general-purpose deep learning architectures, and the specialized and advanced method for this dataset.
3) We evaluate the contribution of nonlocal interactions and rational nonlinearities through ablation studies. Rather than merely validating architectural soundness, these analyses serve to elucidate the nature of aging signals in steroid metabolomics. Specifically, we investigate whether biological age is encoded primarily in individual metabolite abundances or in the complex, coordinated homeostatic networks captured by our interaction-aware modules, thereby linking model performance directly to the systemic dysregulation characteristic of aging.
4) We perform a feature perturbation analysis to demonstrate the interpretability of METRON. This analysis identifies the key metabolic drivers of biological age, revealing that the model's predictive ability is concentrated in a subset of biologically significant metabolites, thereby providing original insights into the aging process.

II. Related Work

A. Biological Age Biomarkers: From Phenotypes to Metabolomics

The pursuit of accurate BA biomarkers [6], [11] has evolved significantly. Early studies relied on phenotypic indicators, such as lung capacity and grip strength [12], [13]. However, these measures often lag behind molecular changes. Consequently, the field has shifted towards omics technologies, including genomics, epigenomics, proteomics, and metabolomics, which reveal complex interactions at a molecular level [5], [14]. In particular, metabolomics studies the intermediate and downstream products of cellular processes [15]. Since metabolomic profiles reflect real-time influences from diet, microbiome, and environment, they are particularly well-suited for capturing the dynamic aging process.

B. Machine Learning and Deep Learning in Aging Research

The complexity of aging necessitates advanced computational models. Traditional machine learning methods like LASSO and ridge regression have been widely applied to omics data [16], [17]. However, their capacity is often limited to linear relationships, potentially overlooking the intricate interactions inherent in biological systems [6]. To overcome these limitations, the community has increasingly adopted nonlinear models. Support vector machines (SVMs) [18] and random forests [19] have demonstrated improved performance. More recently, deep neural networks (DNNs) have gained prominence for their ability to model high-dimensional, nonlinear dynamics [20]. For instance, Mamoshina et al. [21] successfully applied DNNs to identify age-related markers from gene expression data. Despite their powerful fitting capabilities, standard DNNs are susceptible to overfitting and often lack interpretability, which is crucial for biomedical applications [10], [22].

To address the limitations of standard architectures, the community has recently explored more advanced deep learning methods. For example, Transformer-based models have been introduced to capture global dependencies in high-dimensional omics data [23]. Similarly, graph neural networks (GNNs) have been utilized to explicitly model the topological structure of biochemical pathways and metabolite interactions [24]. While these approaches have improved representation capability, they often introduce considerable computational complexity and require massive datasets for training, which are not always available in specific clinical metabolic studies.

C. State-of-the-Art in Steroid-Based BA Estimation

In the specific domain of steroid metabolomics, a key recent study by Wang et al. [9] established a benchmark by quantifying more than 20 steroids using LC-MS/MS and proposing a pathway-based DNN. Their architecture was designed to mirror known biochemical pathways. However, this approach has several limitations that motivate our current work. First, it is not a complete end-to-end framework, relying on extensive statistical preprocessing, including CDF-based scaling and Yeo-Johnson transformation. Second, it requires domain-specific knowledge for weight initialization, which restricts flexibility. Third, the necessity of training separate models for male and female subjects limits its generalizability. These challenges highlight the need for a unified, end-to-end architecture capable of adaptively capturing metabolic interactions, which leads to the proposal of our METRON framework.

D. Kolmogorov-Arnold Networks (KANs)

While multi-layer perceptrons (MLPs) serve as the cornerstone of deep learning, they primarily rely on linear weights followed by fixed node-wise activation functions. Recently, inspired by the Kolmogorov-Arnold representation theorem, Liu et al. [25] introduced Kolmogorov-Arnold Networks (KANs). Unlike MLPs, KANs place learnable univariate activation functions on the edges, typically parametrized as B-splines, eliminating linear weights entirely. This paradigm shift offers superior interpretability and data efficiency, theoretically allowing for faster neural scaling laws compared to MLPs.

However, standard KANs often face computational inefficiencies and numerical instability when scaled to deep architectures or high-dimensional data. To address these limitations, Yang et al. [26] proposed the Kolmogorov-Arnold Transformer (KAT), which introduces the group-rational KAN (GR-KAN). This architecture substitutes the computationally

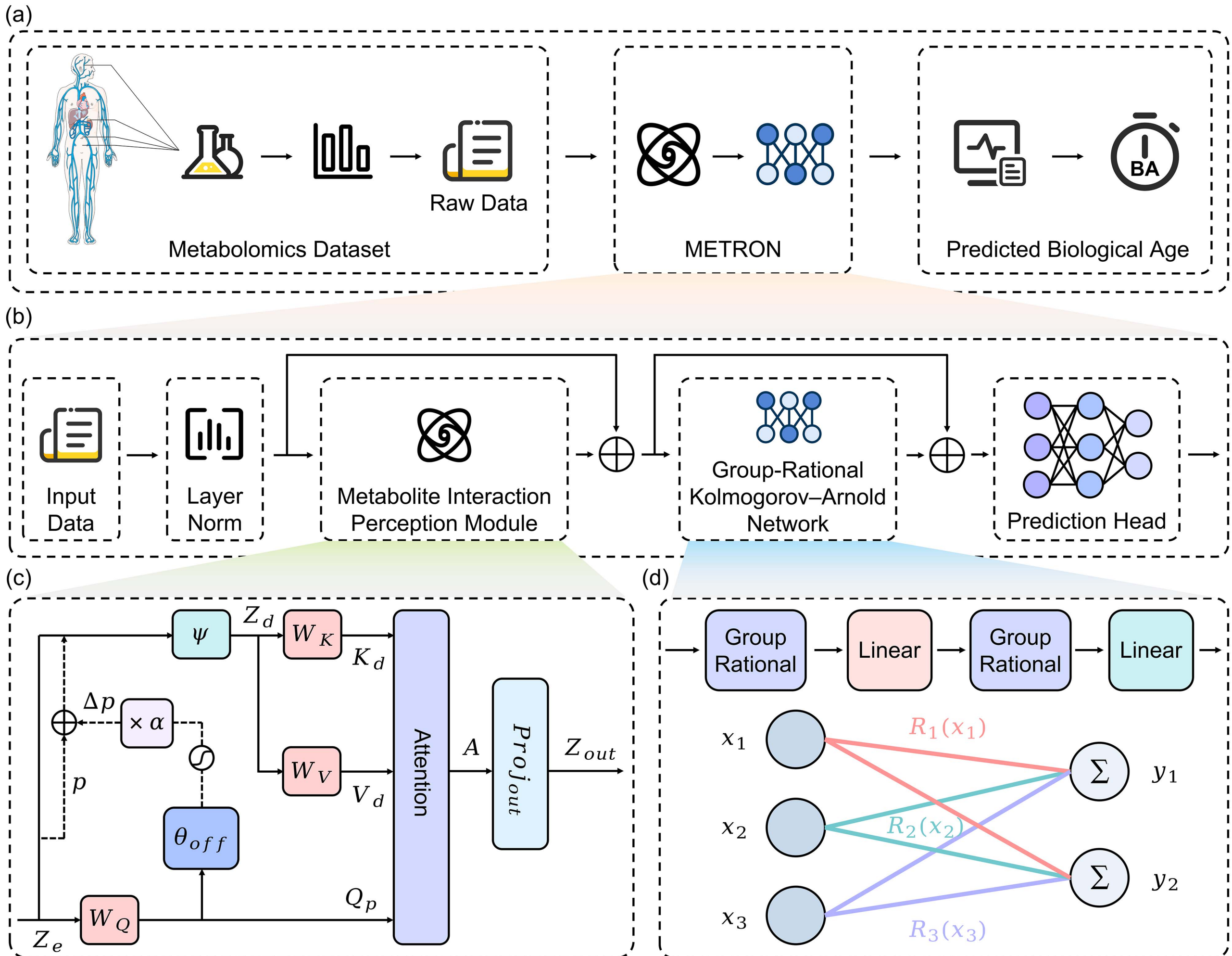

Fig. 1. Proposed Architecture of METRON Framework for Biological Age Estimation. (a) The overall workflow. (b) The detailed architecture of METRON. (c) The internal structure of the Metabolite Interaction Perception Module. (d) The GR-KAN layer. The colors indicate that the parameters of the rational functions are shared within distinct input groups. Illustration created with elements from Servier Medical Art, licensed under CC BY 3.0.

expensive B-splines with rational basis functions and employs group-shared parameters to enhance numerical stability and reduce computational overhead. In this study, we leverage the GR-KAN architecture within our METRON framework. This choice allows us to efficiently model the complex, high-order nonlinear interactions within steroid metabolic pathways by leveraging the mathematical alignment between rational functions and enzyme kinetic laws (e.g., Michaelis-Menten kinetics).

III. Materials and Methods

A. Metabolomics Dataset and Acquisition

The metabolomics dataset employed in this study was derived from a cohort of 148 healthy individuals, originally detailed by Wang et al. [9]. The dataset captures the physiological status of participants through the precise quantification of 22 serum steroid hormones. Consistent with the previous benchmark, the data is partitioned into a training cohort ($n = 98$; 49 females, 49 males; age range: 20–73 years) and an independent validation cohort ($n = 50$; 25 females, 25 males; age range: 40–59 years) used exclusively for final model assessment.

Serum samples were processed using a liquid chromatography-tandem mass spectrometry (LC-MS/MS) protocol to ensure high sensitivity and specificity. Briefly, serum aliquots were spiked with isotope-labeled internal standards for accurate quantification. Protein precipitation was performed using acetonitrile, followed by centrifugation. The supernatant underwent solid-phase extraction (SPE) on a C18 column to isolate the steroid fraction. The purified analytes were separated and quantified via LC-MS/MS operating in multiple reaction monitoring (MRM) mode. This method simultaneously targeted a panel of 22 steroids, including key aging biomarkers such as dehydroepiandrosterone (DHEA), cortisol (COL), 17-hydroxyprogesterone (17-OH-P4), estrone (E1), testosterone (TE), and progesterone (P4).

The resulting raw data consists of high-dimensional vectors representing the serum concentrations of these 22 steroids and demographic data. Unlike traditional phenotypic markers, these continuous variables capture nonlinear metabolic fluctuations. To provide insight into the data distribution and scale, a representative subset of the raw input data (quantified concentrations) is presented in Table I. Each sample serves

TABLE I
REPRESENTATIVE SAMPLES OF RAW STEROID CONCENTRATIONS FROM THE DATASET.

ID	Group	Sex	Age	DHEA	COL	17-OH-P4	TE	COR	AT	DHT	THB	THS	DOC
9525	Training	Female	60	3.5856	70.3889	0.1704	0.1631	9.5722	0.1061	0.1082	0.1297	0.0113	0.0645
3977	Training	Male	21	6.9770	25.4589	1.1929	0.6263	5.7700	0.0640	0.1599	0.0560	0.0140	0.0497
4002	Training	Female	37	8.4813	72.8381	0.9168	0.0654	7.8009	0.0736	0.0886	0.5142	0.0180	0.0387
3986	Training	Male	39	7.8434	88.3917	0.3071	2.1974	12.8666	0.1770	0.2246	0.3412	0.0194	0.0511
9548	Validation	Female	52	5.9774	75.6872	0.1397	0.1292	11.7352	0.1736	0.1138	0.0950	0.0042	0.0597
2099	Validation	Male	45	3.2502	38.8414	0.2848	1.8195	7.6112	0.1409	0.1656	0.0511	0.0169	0.0366

Note: This table only presents a subset of the raw input features used in the study.

as the input to our framework, where specific steroid levels reflect the underlying hypothalamic-pituitary-adrenal (HPA) and gonadal axis dynamics.

B. Framework Overview

The proposed METRON framework is designed to predict biological age from metabolomics data. As illustrated in Fig. 1(a), the raw metabolomics data is fed into our model, which processes the complex relationships between metabolites to generate a scalar prediction of an individual's biological age. The core architecture of our model, depicted in Fig. 1(b), is structured as a sequential cascade of specialized modules. The input data, representing the concentrations of various metabolites, first passes through a Layer Normalization step to stabilize the feature distribution. The normalized data is then processed by two principal components: a Metabolite Interaction Perception Module and a Group-Rational Kolmogorov-Arnold Network, which are interconnected via residual connections. The Metabolite Interaction Perception Module is the first key component of our model, which is designed to capture the complex, non-local interactions between different metabolites. As shown in Fig. 1(c), unlike standard self-attention mechanisms that operate over a fixed set of input features, this module dynamically learns sampling offsets ($\Delta \boldsymbol{p}$). These offsets guide the attention mechanism to focus on the most salient metabolite relationships for the age prediction task, regardless of their position in the input feature vector. This deformable sampling allows the model to build a more flexible and robust understanding of the underlying metabolic interplay. Following the Metabolite Interaction Perception Module, the features are transformed by a feature extraction backbone. Instead of using conventional Multi-Layer Perceptrons (MLPs), we employ a Group-Rational Kolmogorov-Arnold Network (GR-KAN). As illustrated in Fig. 1(d), a GR-KAN layer replaces fixed activation functions (e.g., ReLU) with learnable, group-wise rational functions ($R(x)$). These functions are applied to the input features before a subsequent linear transformation. Stacking these GR-KAN and linear layers provides superior expressive power compared to standard MLPs, allowing the model to learn more intricate and accurate representations of the data. The group structure, where rational function parameters are shared across groups of neurons, enhances computational efficiency. The final stage of the framework is a simple yet effective prediction head. It consists of a stack of fully connected layers that take the rich feature embedding from the GR-KAN block and regress it to a single continuous value, which corresponds to the predicted biological age. Residual connections are utilized throughout the model to ensure effective gradient flow during training and prevent degradation.

C. Metabolite Interaction Perception Module

To effectively capture the complex and potentially long-range dependencies between metabolites, we introduce the Metabolite Interaction Perception Module. The design of this module is inspired by the deformable attention mechanism, which has demonstrated significant success in computer vision and time series analysis [27]. In contrast to standard self-attention, which computes dense interactions across all input features, our module adaptively focuses on a small, salient set of key metabolites, enhancing both model efficiency and performance.

Let the input to the module be the normalized feature matrix $\boldsymbol{Z}_{\mathrm{e}}$. The core of the module is to augment the standard attention mechanism with a data-dependent offset learning process. First, a set of reference points $\boldsymbol{p}$ is defined, representing the initial grid of metabolite features. A query projection $\boldsymbol{Q}_p$ is computed from the input features $\boldsymbol{Z}_{\mathrm{e}}$. This query is then fed into a lightweight sub-network, denoted as θ_{off}, to generate a set of sampling offsets $\Delta \boldsymbol{p}$. This process can be formulated as:

$$\Delta \boldsymbol{p} = \alpha \tanh(\theta_{\mathrm{off}}(\boldsymbol{Q}_p)) \tag{1}$$

Here, θ_{off} is typically a simple neural network, such as a linear layer. The tanh function constrains the magnitude of the learned offsets, ensuring that the sampling locations do not stray too far from their reference points. The scalar α is a learnable amplitude that controls the maximum range of the deformation, providing an additional degree of freedom.

The computed offsets $\Delta \boldsymbol{p}$ are then added to the original reference points $\boldsymbol{p}$ to determine the new sampling locations, $\boldsymbol{p} + \Delta \boldsymbol{p}$. Since these new locations are continuous and do not necessarily align with the discrete grid of input features, we employ an interpolation function, ψ, to sample the features from $\boldsymbol{Z}_{\mathrm{e}}$ at these deformed locations. This creates the deformed key ($\boldsymbol{K}_{\mathrm{d}}$) and value ($\boldsymbol{V}_{\mathrm{d}}$) feature maps:

$$\boldsymbol{Z}_{\mathrm{d}} = \psi(\boldsymbol{Z}_{\mathrm{e}}; \boldsymbol{p} + \Delta \boldsymbol{p}) \tag{2}$$

where $\boldsymbol{Z}_{\mathrm{d}}$ represents the features sampled from the deformed locations. The deformed keys and values are then obtained through linear projections $\boldsymbol{K}_{\mathrm{d}} = \boldsymbol{Z}_{\mathrm{d}} \boldsymbol{W}_{\mathrm{K}}$ and $\boldsymbol{V}_{\mathrm{d}} = \boldsymbol{Z}_{\mathrm{d}} \boldsymbol{W}_{\mathrm{V}}$, where $\boldsymbol{W}_{\mathrm{K}}$ and $\boldsymbol{W}_{\mathrm{V}}$ are learnable weight matrices. Similarly, a query projection $\boldsymbol{Q}_p$ is generated from the original input features $\boldsymbol{Q}_p = \boldsymbol{Z}_{\mathrm{e}} \boldsymbol{W}_{\mathrm{Q}}$.

Finally, a standard dot-product attention operation is performed using the query $\boldsymbol{Q}_p$ and the deformed keys $\boldsymbol{K}_{\mathrm{d}}$ and values $\boldsymbol{V}_{\mathrm{d}}$:

$$\mathrm{Attention}(\boldsymbol{Q}_p, \boldsymbol{K}_{\mathrm{d}}, \boldsymbol{V}_{\mathrm{d}}) = \mathrm{softmax}\left(\frac{\boldsymbol{Q}_p \boldsymbol{K}_{\mathrm{d}}^{\top}}{\sqrt{d_k}}\right)\boldsymbol{V}_{\mathrm{d}} \quad (3)$$

The output of the attention mechanism is then passed through a final projection layer, $\mathrm{Proj}_{\mathrm{out}}$, to yield the module's output $\boldsymbol{Z}_{\mathrm{out}}$. By learning where to attend, the Metabolite Interaction Perception Module can flexibly and efficiently model critical relationships within the metabolome for the task of biological age prediction.

D. Group-Rational Kolmogorov-Arnold Network

Inspired by the previous work [26], we utilized the Group-Rational Kolmogorov-Arnold Network (GR-KAN) as the core of our model's feature extraction capability. Unlike traditional Multi-Layer Perceptrons (MLPs) that use fixed, non-learnable activation functions such as ReLU, the GR-KAN leverages learnable activation functions to achieve superior expressive power, making it highly suitable for modeling the intricate relationships in metabolomics data. The GR-KAN is an efficient and scalable variant of the original Kolmogorov-Arnold Network (KAN) [25]. It addresses the primary challenges of the KAN, including computational cost and slow performance on GPUs, through two key innovations: the use of rational functions as a base and a parameter-sharing strategy.

Instead of the B-spline functions used in the original KAN, the GR-KAN employs rational functions, which are ratios of two polynomials, $P(x)$ and $Q(x)$. These functions are computationally efficient and well-suited for parallel processing. Crucially, the choice of rational functions provides a biological inductive bias specifically suited for metabolic data. The fundamental dynamics of enzyme kinetics, which govern metabolic flux, are classically described by the Michaelis-Menten equation [28], [29] ($v = \frac{V_{max}[S]}{K_m+[S]}$) or the Hill equation [30] for cooperative binding. Mathematically, these laws are inherent rational functions (ratios of polynomials). This stands in contrast to other learnable activations such as B-splines (piecewise polynomials) or SIRENs (periodic sine waves). While SIRENs are effective for high-frequency signals, metabolic dose-response curves typically exhibit monotonic and saturation behaviors rather than periodicity [29], [31]. Similarly, while B-splines are universal approximators, rational functions naturally align with the underlying physicochemical laws of saturation kinetics, allowing the network to more efficiently capture the nonlinear dynamics of steroidogenesis. For each edge in the network, the learnable activation $\phi(x)$ is defined as:

$$\phi(x) = wF(x) = w\frac{P(x)}{Q(x)} = w\frac{a_0 + a_1 x + \cdots + a_m x^m}{b_0 + b_1 x + \cdots + b_n x^n} \quad (4)$$

Here, the polynomial coefficients a_i and b_j, along with the scaling factor w, are learnable parameters that are optimized during training via backpropagation.

To reduce the number of parameters and the computational load, the GR-KAN introduces a grouping strategy. A conventional KAN requires a unique learnable function for every input-output pair, leading to a quadratic increase in parameters. In contrast, the GR-KAN divides the d_{in} input channels into g groups. Within each group, the parameters of the rational function $F(x)$ (i.e., the coefficients a_i and b_j) are shared across all edges originating from that group. However, each individual edge retains a unique learnable scalar weight w, preserving the flexibility of the model. The operation of a GR-KAN layer can be understood as a two-stage process. First, a group-wise rational function is applied to the input features. Second, a standard linear transformation is applied to the output of the functions. This can be expressed in matrix form as:

$$\text{GR-KAN}(\boldsymbol{X}) = \boldsymbol{W}\boldsymbol{F}(\boldsymbol{X}) \quad (5)$$

where $\boldsymbol{X}$ is the input feature vector, $\boldsymbol{F}(\boldsymbol{X})$ is the vector resulting from the application of the appropriate group-wise rational function to each element of $\boldsymbol{X}$, and $\boldsymbol{W}$ is the weight matrix of the subsequent linear layer. This structure effectively functions as a specialized MLP where the activation is both learnable and precedes the linear layer.

IV. Results

A. Evaluation Metrics

To quantitatively evaluate the performance of our model, a comprehensive set of metrics was employed. First, we utilized standard regression metrics, including the mean absolute error (MAE), the mean squared error (MSE), the root mean squared error (RMSE), the mean absolute percentage error (MAPE), and the Pearson correlation coefficient (PCC). To further assess the model's discriminative ability in ranking evaluations, we incorporated the concordance index (C-index). In addition to these standard measures, we adopted the weighted symmetric arctangent loss (WSATL) [9]. This metric addresses the challenge of increasing aging heterogeneity, where prediction variance tends to increase with age. The function balances prediction accuracy by symmetrically penalizing disproportionate errors while handling high and low biases across chronological age ranges. Finally, to quantify the dispersion of predictions relative to the identity line in scatter plot visualizations, the root mean squared offset deviation (RMSOD) was calculated. Detailed mathematical formulations for all metrics are provided in the supplementary material.

B. Comparative Performance Analysis

To evaluate the performance of our proposed model, we conducted a comprehensive comparative analysis against a wide range of established methods. We began with pathway-based DNN, a method specifically designed for this metabolomics dataset. Consistent with previous literature, we evaluated a suite of machine learning algorithms renowned for their efficacy on tabular data, which appear frequently in previous biological age prediction studies [43], [44]. These include LASSO, Ridge Regression, Support Vector Regression (SVR), K-Nearest Neighbors (KNN), Random Forest, Gradient Boosting Decision Trees (GBDT), XGBoost, LightGBM, and MLP. Furthermore, given the limited number of publicly available methods tailored to this specific metabolomics dataset and to represent advanced

TABLE II
RESULTS OF COMPARATIVE PERFORMANCE ANALYSIS. BOLDED VALUES INDICATE THE BEST PERFORMANCE FOR EACH METRIC.

Category	Method	WSATL	MAE	MSE	RMSE	MAPE	PCC
Baseline ML Methods	LASSO [32]	0.0023	5.8854	61.5532	7.8455	11.7755	0.4069
	Ridge Regression [33]	0.0039	8.3298	139.6466	11.8172	17.2657	0.2670
	Support Vector Regression [34]	0.0024	6.3249	54.2765	7.3672	12.2785	0.4459
	K-Nearest Neighbors Regression [35]	0.0036	9.1979	116.1739	10.7784	18.5402	0.4761
	Random Forest [36]	0.0022	6.3027	56.5630	7.5208	13.4341	0.3611
	Gradient Boosting Decision Tree [37]	0.0028	9.0513	123.1941	11.0992	19.2619	0.0895
	XGBoost [38]	0.0026	8.1683	99.4563	9.9727	17.2748	0.1609
	LightGBM [39]	0.0022	6.7826	71.8470	8.4762	14.3427	0.3839
Deep Learning Methods	MLP [40]	0.0032	8.5871	106.5143	10.3205	17.4893	0.4524
	TextCNN [41]	0.0025	7.1588	79.9375	8.9407	14.2001	0.4705
	FT-Transformer [42]	0.0037	8.6791	114.9132	10.7197	17.7244	0.3855
Specialized Methods	Pathway-based DNN [9]	0.0038	9.0980	132.0721	11.4923	18.3671	0.4585
	METRON (Ours)	**0.0018**	**4.4656**	**36.7943**	**6.0658**	**9.0278**	**0.6036**

deep learning architectures, we also incorporated TextCNN, originally designed for sequence data, and FT-Transformer, a leading-edge model specifically developed for tabular data. To ensure a fair comparison, all models were trained and evaluated using identical experimental settings.

The results are summarized in Table II. Our model consistently outperformed all other methods across all six evaluation metrics. It achieved the lowest WSATL (0.0018), MAE (4.4656), MSE (36.7943), RMSE (6.0658), and MAPE (9.0278), while also attaining the highest PCC (0.6036). These results indicate a substantial improvement in prediction accuracy and reliability. Among the base methods, several models showed competitive performance relative to one another. LASSO achieved the lowest MAE of 5.8854, while K-NN Regression yielded the highest PCC of 0.4761. However, our model's performance represents a leap forward. For instance, our model's MAE of 4.4656 is approximately 24% lower than that of LASSO. Furthermore, its PCC of 0.6036 is higher than any base method. Among the deep learning methods, TextCNN showed the strongest performance with an MAE of 7.1588 and a PCC of 0.4705. Nevertheless, our model demonstrated lower error rates compared to these deep learning methods, including the FT-Transformer. The performance gap underscores the effectiveness of our specialized architecture compared to more generalized deep learning approaches.

Compared with the specialized method, our model shows a clear advantage over the pathway-based DNN. Our model's scores were superior across all metrics, with an MAE of 4.4656 compared to 9.098 for the pathway-based DNN, and a PCC of 0.6036 versus 0.4585. These results underscore the enhanced predictive capability of our proposed architecture for biological age prediction using metabolomics data, establishing its effectiveness in capturing the complex associations among metabolites.

C. Evaluation of Concordance and Ranking Performance

To further validate the robustness of METRON, we employed the concordance index (C-index) as a key evaluation metric. Unlike MAE or MSE, which quantify absolute prediction errors, the C-index measures the model's ability to correctly rank the relative biological ages of individuals. This capability is critical for distinguishing between accelerated and decelerated aging phenotypes in clinical applications.

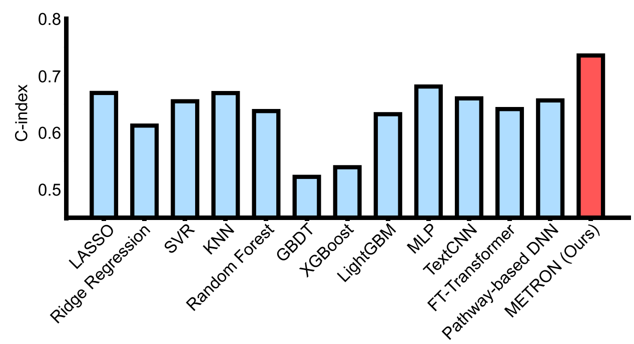

Fig. 2. Performance comparison based on the concordance index (C-index).

As illustrated in Fig. 2, METRON achieved the highest C-index of 0.7353, outperforming all baseline methods. Among the deep learning competitors, MLP showed the second-best performance (0.6808), followed by TextCNN (0.6600) and pathway-based DNN (0.6564). In terms of traditional machine learning, linear models like LASSO (0.6695) and distance-based methods like KNN (0.6693) demonstrated reasonable ranking abilities. However, ensemble tree-based models, including GBDT (0.5222) and XGBoost (0.5391), performed poorly on this dataset, yielding results close to random guessing (0.5). These results confirm that METRON captures the global aging trend more effectively than other approaches. The high C-index indicates that our model can reliably identify which individuals are physiologically older, providing a trustworthy tool for aging assessment.

D. Ablation Study

To investigate the individual contributions of the core modules and their correspondence to biological mechanisms, we conducted an ablation study by removing the metabolite interaction perception module (MIPM) or the group-rational Kolmogorov-Arnold Network (GR-KAN).

The results (Fig. 3) reveal that our full model outperforms the ablated variants, confirming that both modules capture essential

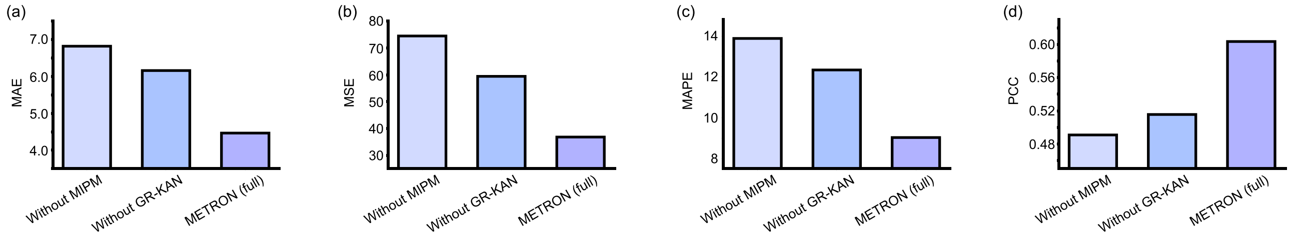

Fig. 3. Performance comparison of the full model against two ablated versions on the test set. The panels show the results for four different metrics: (a) MAE; (b) MSE; (c) MAPE; and (d) PCC.

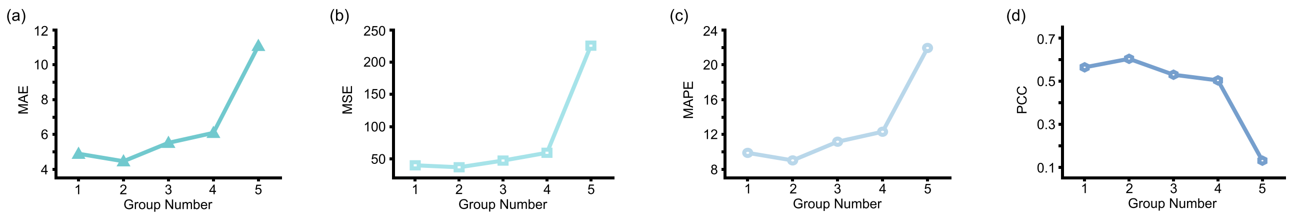

Fig. 4. Parameter analysis for the number of groups in the GR-KAN. The panels correspond to four different metrics: (a) MAE; (b) MSE; (c) MAPE; and (d) PCC. The model achieves its optimal performance across all metrics when g is set to 2.

biological properties of the data. The removal of the MIPM caused the most severe performance degradation, with MAE increasing by 52.8% (Fig. 3(a)) and PCC dropping to 0.4908 (Fig. 3(d)). The MIPM is designed to model nonlocal dependencies. Biologically, this drastic drop implies that biological age is not merely reflected by the decline of individual steroids (e.g., DHEA levels alone), but is encoded in the dysregulation of the inter-metabolite coordination network, which is the loss of homeostasis. The model's reliance on MIPM validates the hypothesis that aging is a systemic failure of metabolic interaction patterns rather than isolated marker changes.

Similarly, removing the GR-KAN led to a notable decline (MAE rose to 6.1618). As discussed in Section III-D, the rational functions in GR-KAN mathematically align with enzyme kinetic laws (e.g., Michaelis-Menten kinetics), which inherently involve ratios of polynomials. The superior performance of GR-KAN over standard feature extractors suggests that capturing these specific saturation and regulatory dynamics inherent to steroidogenesis is crucial for accurate modeling. The ablation results thus confirm that the model's architecture succeeds by mirroring the underlying physicochemical laws of the metabolic system.

E. Parameter Analysis of Group-Rational Kolmogorov-Arnold Network

We investigated the critical role of the grouping parameter (g) in GR-KAN. This parameter controls the extent of parameter sharing among learnable activation functions, modeling the diversity of enzymatic kinetics within the network.

As shown in Fig. 4, the optimal performance is achieved at $g = 2$ (MAE $= 4.4656$). Setting $g = 1$ enforces global parameter sharing, implying that all metabolic transformations follow identical kinetic curves. The higher error at $g = 1$ suggests this assumption is too restrictive to capture the diversity of steroid enzymes (e.g., hydroxylation vs. dehydrogenation). Conversely, increasing g beyond 2 led to sharp performance degradation (e.g., PCC dropped to 0.1307 at $g = 5$). While this reflects statistical overfitting, biologically, it also suggests that the metabolic signal is not comprised of independent, disjointed processes. The optimality of a small group size ($g = 2$) implies a functional modularity in the steroid pathway, where metabolites function in small, tightly regulated groups sharing similar kinetic properties. Thus, $g = 2$ strikes a balance that likely reflects the true complexity of the underlying enzymatic clusters.

F. Visualization of Predictive Performance

To provide a qualitative and intuitive assessment of the performance of our model against other models, we visualized the prediction results for different models on the test set. Fig. 5 presents scatter plots of the predicted biological age against the chronological age for each model. The dashed line in each plot represents the line of perfect prediction ($y = x$), where predicted age equals chronological age.

A common trend observed across the baseline models (Fig. 5(a)–(k)) is the dispersion of predicted values. Notably, regarding the interpretation of clustering patterns, a distinction must be made between visual compactness and orthogonal accuracy. For instance, the predictions of Random Forest (Fig. 5(d)) appear tightly clustered in a narrow range (approximately 40–50 years). However, this visual compactness is indicative of a regression to the mean effect, where the model fails to capture the full variance of the aging process, resulting in a higher RMSOD (5.3180). This indicates that while the points are close to each other, they are statistically further from the ground truth line compared to our method. In contrast, our model (Fig. 5(l)) achieves the lowest RMSOD (4.2892) among all evaluated methods. Although the data points cover a wider range along the diagonal, reflecting a better capture of the natural physiological variance, they adhere more strictly to the $y = x$ line. This

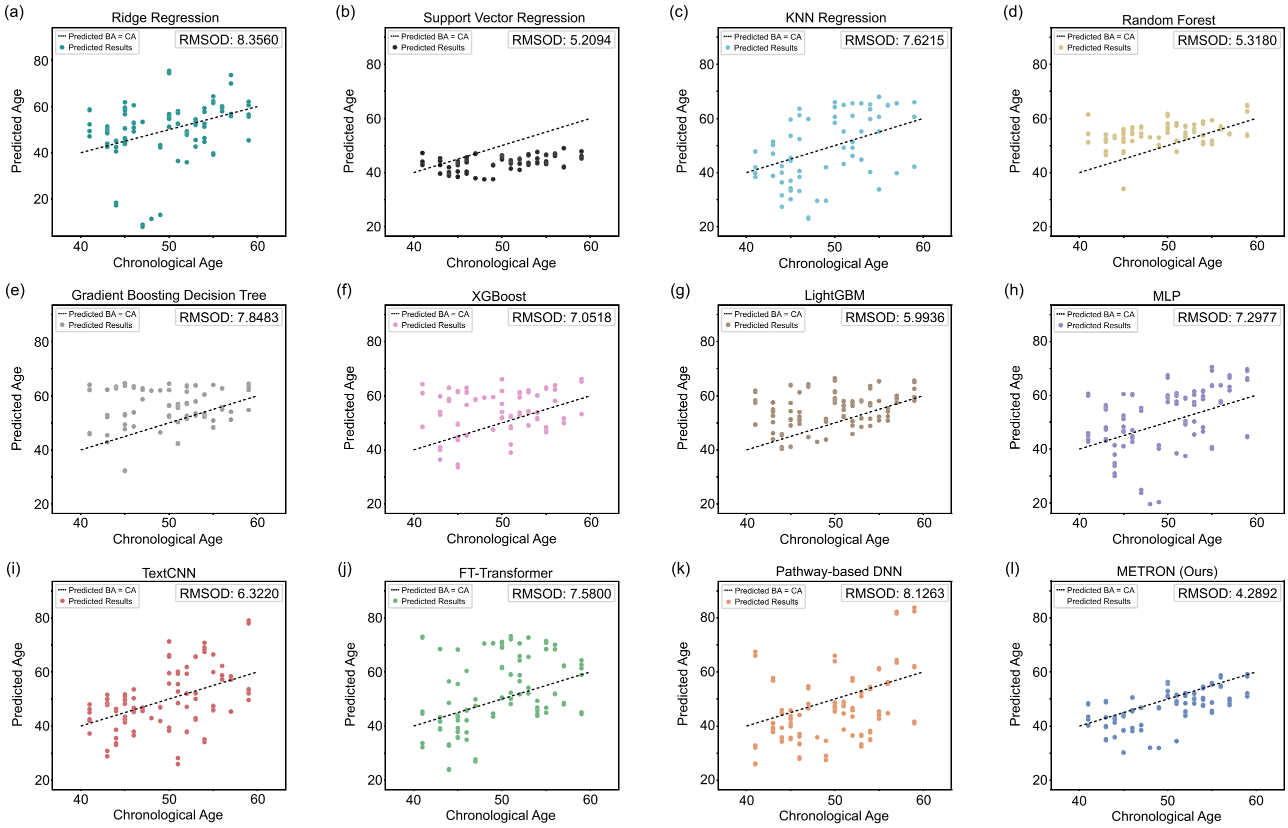

Fig. 5. Visualization of predictive performance for different models on the test set. The dashed line indicates perfect prediction ($y = x$). To quantify the deviation from this ideal line, RMSOD is calculated for each model (lower is better).

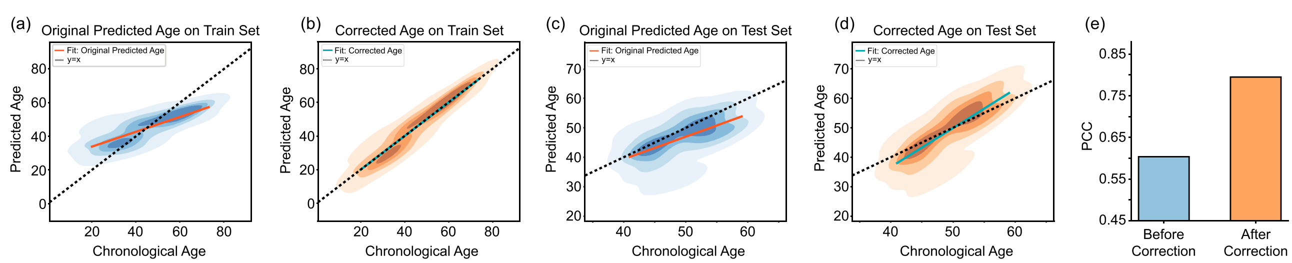

Fig. 6. Statistical correction of systematic age prediction bias. (a), (c) Two-dimensional density plots of predicted age versus chronological age on the training (a) and test (c) sets before correction. (b, d) Corresponding plots for the training (b) and test (d) sets after statistical correction. Solid colored lines indicate the linear regression fit. (e) Bar chart quantifying the increase in the PCC on the test set after correction.

lower RMSOD demonstrates a greater concordance between the predicted biological age and the actual chronological age, confirming that METRON offers higher precision and reduces overall prediction error effectively.

G. Age-Bias Correction

While METRON demonstrates strong feature extraction capabilities, the raw output exhibited a systematic compression of the predicted age range, overestimating younger individuals and underestimating older ones. It is crucial to note that this is a well-documented statistical phenomenon known as regression to the mean (RTM) [45], [46], which is intrinsic to regression tasks on biological datasets and independent of the underlying model architecture. This systematic linear offset has been ubiquitously observed in state-of-the-art biological age prediction frameworks, including recent benchmarks [9], [44]. Therefore, applying a statistical correction is a standard calibration procedure to ensure the predicted values are biologically interpretable.

To eliminate this systematic bias and ensure a fair evaluation, we applied the standard linear correction method described in [45]. The specific effects are illustrated in Fig. 6. Prior to correction, the predictions (Fig. 6(a), (c)) displayed the characteristic RTM pattern with a regression slope shallower than the identity line ($y = x$). However, the model successfully captured the relative ordering of samples, indicating that the core MIPM and GR-KAN modules effectively learned the nonlinear aging patterns. Following the correction, the predictions are properly

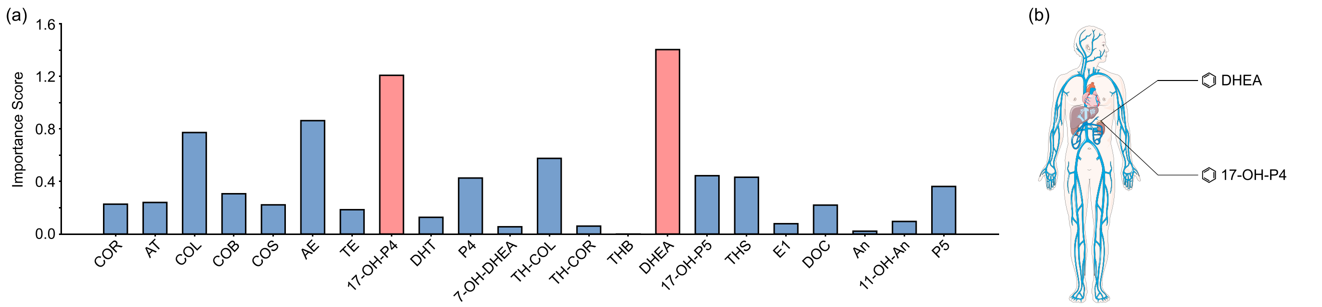

Fig. 7. Hierarchy of metabolite importance for biological age prediction. (a) The bar plot displays the permutation importance score for each individual metabolite. The importance score is defined as the increase in model MAE after values of a metabolite were permuted. Scores below zero, assumed to be random noise, were set to zero. (b) Our model identifies DHEA and 17-OH-P4 as important metabolites. Illustration created with elements from Servier Medical Art, licensed under CC BY 3.0.

re-scaled to align with chronological age (Fig. 6(b), (d)), tightening the distribution around the diagonal. Quantitatively, this calibration improved the PCC on the test set from 0.6036 to 0.7950 (Fig. 6(e)). This enhancement confirms that the scatter observed in raw predictions was primarily a result of systematic scaling bias rather than random error, and the corrected METRON achieves superior alignment with the ground truth.

H. Permutation Analysis Reveals Key Metabolic Drivers of Age Prediction

To identify the most influential metabolites driving the biological age predictions, we employed a permutation feature importance (PFI) analysis. This method assesses the contribution of each metabolite by measuring the drop in model performance when the information of that metabolite is disrupted. Specifically, for each of the 22 metabolites, we permuted (shuffled) its values across all samples in the test set. All other features were kept unchanged. This perturbed dataset was then passed to our trained model to generate new predictions. The importance score for that metabolite was defined as the increase in MAE compared to the MAE from the unaltered test set. To ensure stable and robust estimates, this random permutation process was repeated 10 times for each metabolite. The final importance score was taken as the average of these 10 trials. Following this process, any metabolite with a negative average score, which can arise from random chance when a feature is uninformative, was considered to have no predictive value, and its importance was set to zero.

The results of this analysis revealed a hierarchy of metabolic importance (Fig. 7(a)). The predictive ability of our model was not distributed evenly. Instead, it was concentrated in a subset of metabolites.

The model was exceptionally sensitive to DHEA, which emerged as the most critical metabolite. Its permutation resulted in a substantial performance drop. This finding aligns with established aging biology. Dehydroepiandrosterone (DHEA) is an adrenal steroid hormone and represents one of the most well-documented biomarkers of human aging [47]. Its circulating levels, which are often measured as its sulfated form, DHEAS, are known to peak in young adulthood and then decline dramatically with age, a phenomenon termed adrenopause [48]. Consequently, low DHEA(S) levels have been consistently associated with age-related frailty, morbidity, and all-cause mortality [49]. The fact that our model identified DHEA as the top predictor strongly validates its ability to capture a genuine and fundamental signal of the human aging process.

More striking, however, was the identification of 17-OH-P4 (17-hydroxyprogesterone) as the second most important metabolite, yielding an importance score of 1.2082. This finding is particularly novel. 17-OH-P4 has been historically overlooked in aging research and is primarily utilized in clinical practice as the gold-standard diagnostic biomarker for congenital adrenal hyperplasia (CAH), a genetic disorder of cortisol synthesis [50]. Our model's discovery of its high importance in predicting biological age is, to our knowledge, unexpected. As 17-OH-P4 is a direct upstream precursor to cortisol, this result may indicate that subtle, age-related changes in the flux of the adrenal steroidogenic pathway (part of the HPA axis) are a critical and underappreciated component of aging.

Following these two key adrenal steroids, a secondary group of metabolites, including AE (0.8633), COL (0.7727), TH-COL (0.5759), 17-OH-P5 (0.4441), THS (0.4323), P4 (0.4264), P5 (0.3622), and COB (0.3066) showed a modest but distinct contribution. Beyond this top-tier group, the contributions of the remaining metabolites dropped off. The vast majority displayed small importance scores, with scores at or below 0.23. While many of these, such as DHT (0.1272) and 11-OH-An (0.0957), retained a positive score, metabolite THB got an importance score of zero after our thresholding. This finding suggests that the high accuracy of our model is not derived from uniform information across the metabolome but is instead reliant on a few high-importance metabolic markers.

V. Discussion

In this study, we presented METRON, a novel deep learning framework designed to predict biological age from steroid metabolomics data. By integrating a Metabolite Interaction Perception Module (MIPM) with a Group-Rational Kolmogorov-Arnold Network (GR-KAN), METRON effectively addresses the challenges of high dimensionality and nonlinearity inherent in omics data. The comprehensive evaluation demonstrates that METRON not only achieves superior predictive accuracy compared to classical machine learning and advanced deep learning methods (such as pathway-based DNN) but also provides interpretable insights into the metabolic drivers of aging.

The promising performance of METRON can be attributed to its architecture tailored for biological data. Traditional methods such as LASSO often fail in capturing the intricate and nonlinear interactions between metabolites. While standard deep learning models (e.g., MLPs and Transformers) offer nonlinear capabilities, they frequently suffer from overfitting on smaller biological datasets or lack specific inductive biases for metabolic networks. The proposed MIPM allows the model to dynamically focus on metabolite interactions, akin to how biological pathways operate through specific, often nonlocal, regulatory feedbacks. Furthermore, the GR-KAN replaces the fixed activation functions of conventional neural networks with learnable rational functions. This design enhances the expressive ability, enabling it to model the complex, heterogeneous trajectories of physiological aging more accurately than standard architectures, as evidenced by our ablation studies.

Beyond accurate prediction, interpretability is equally important. Our feature perturbation analysis reaffirmed the well-established role of Dehydroepiandrosterone (DHEA) as a primary driver of biological aging, validating the molecular relevance of METRON. DHEA levels are known to decline consistently with age, serving as a reliable benchmark for aging research. More importantly, METRON identified 17-hydroxyprogesterone (17-OH-P4) as an important predictor, a finding that offers novel biological insights. Historically, 17-OH-P4 is primarily viewed as a diagnostic marker for congenital adrenal hyperplasia and has received limited attention in the context of general aging. Its emergence as an important predictor in our model suggests that subtle, age-related alterations in the flux of the adrenal steroidogenic pathway may be an underappreciated component of the aging process. Since 17-OH-P4 is a direct precursor to cortisol, this finding points towards the importance of the hypothalamic-pituitary-adrenal (HPA) axis dynamics in maintaining homeostatic balance during aging. This demonstrates the capacity of our deep learning framework to not only replicate known biology but also to uncover potential new targets for gerontological research.

VI. Limitations

Despite the promising results and biological insights, several limitations of this study should be acknowledged. First, the sample size, while sufficient to demonstrate the efficacy of our method compared to baselines, remains relatively small for deep learning applications. Although the robustness of METRON suggests it can learn from limited data well, validation on large-scale, multi-center cohorts is necessary to confirm its generalizability across diverse populations. Second, the current study design is cross-sectional. While METRON can predict biological age based on a snapshot of metabolic data, longitudinal data is required to disentangle causality from correlation. Specifically, further research is needed to verify whether the identified metabolic changes, such as the fluctuations in 17-OH-P4, are drivers of the aging process or merely downstream byproducts of it. Future work will aim to address these constraints by validating the model on longitudinal datasets to track individual aging trajectories over time. Additionally, we plan to expand the METRON framework to integrate multi-omics data, such as proteomics and transcriptomics. This multi-modal approach will facilitate the construction of a more holistic view of the aging process and allow for the exploration of the clinical potential of these metabolic biomarkers in personalized medicine.

VII. Conclusion

We introduced METRON, a deep learning framework designed to estimate biological ages from steroid metabolomics. By integrating the Metabolite Interaction Perception Module (MIPM) with the Group-Rational Kolmogorov-Arnold Network, METRON effectively models the complex and nonlinear dependencies inherent in metabolic data. METRON demonstrates promising performance compared to existing machine learning baselines, deep learning architectures, and the specialized method. Beyond prediction performance, our framework offers explainable interpretability. It successfully identified the aging markers like DHEA, validating its biological robustness. More importantly, it uncovered 17-OH-P4 as a novel and significant predictor of aging. This finding highlights the potential role of the adrenal steroidogenic pathway and HPA axis dynamics in the aging process; it is a connection previously underappreciated in this context. Collectively, these results establish METRON as a useful tool for metabolomics-based aging estimation. Our approach not only provides a precise method for biological age estimation but also demonstrates how interpretable deep learning can reveal the underlying mechanisms. This paves the way for future applications in personalized medicine, facilitating the monitoring of health trajectories and the assessment of anti-aging interventions.

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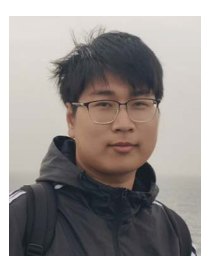

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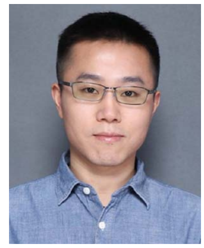

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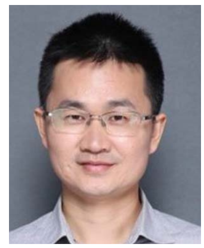

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